INSTITUTE OF TECHNOLOGY OF CAMBODIA

FACULTY OF GEO-RESOURCES AND GEOTECHNICAL ENGINEERING

INTERNSHIP REPORT

TOPIC: Fossil and Stratigraphy Identification to Study Paleo-environment at Mondulkiri and Ratanakiri provinces.

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1. INTRODUCTION

1.1. BACKGROUND STUDY

The term of paleo-environment refers to an environment that existed during the geologic past age. It an encompasses the condition and surrounding in which ancient organisms lived, many factors like: temperature, water depth, energy and chemical composition. Understanding habitats, ecosystems, climates of the past by the evidence from fossils and geologic record.

A fossil is any preserved remains, impression, trace of any living thing from a past geologic age. Fossil are formed when a living organism died and it quickly buried by sediment such as Sedimentary rock and metamorphic rock. Mostly fossils are contained in Carbonate rock. Types of fossils which are Trace fossil, Mold fossil, Cast fossil and Petrified fossil.

Paleoecology, the study of the past environments, is a rapidly changing in a field that involves the integration and synthesis of both botanical and geological information. In recent years there has been a concerted effort by many paleobotanists to understand the paleo- environment of fossil land plants more completely.

Depositional environment is referring to the part or place where sediments deposited. The characterization of depositional environment is important to observed, origin and history of the sediments, age of the rock, past climate, geography and evolution of life earth (Bonewits, 2012) which is provide the clue for better understanding about the relationship between limestone properties and depositional environment by the clue of fossil.

Understanding and interpretation the sedimentological nature of the fossil assemblage, weather based on megafossils and microfossils, is the one of several aspect required in determining the diversification of plants through geologic time. In Mondulkiri occur Ammonite types which are form in Mesozoic era, Jurassic period. Ratanakiri contain petrified wood which is type of palm plant.

1.2. LOCATION OF STUDY AREA

The location of this study area is situated in northeastern of Cambodia. It is named as Sre pork forest WGS1984(765847-1437406), where located in Mondulkiri province, and takes around 521 kilo meters in distance from Phnom Penh. Were mostly occur fossil type Ammonite. And other one location was located near Mondulkiri also it's called Ratanakiri province, which is in Lom Phat WGS1984(720755-1482844), it takes 588 kilometers in distance from Phnom Penh. There is contain of Petrified wood.

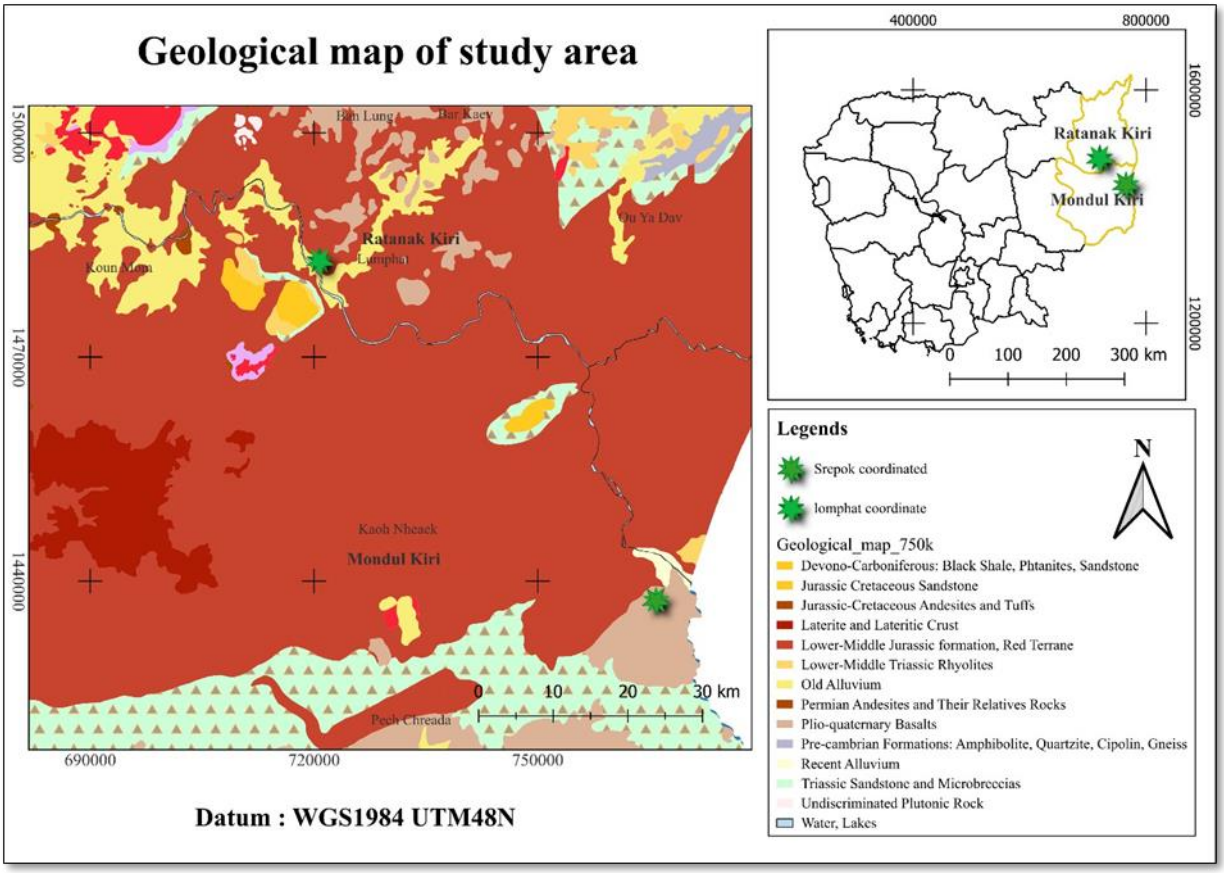


Figure 1-1. Geological map of study area. The green stars are showing at Sre pok forest and Lom phat.

1.3. PROBLEM STATEMENT

Fossil and Petrified wood from one period of geologic time is different in size and shape, and abundance from those of time period. Nevertheless, the characteristic of fossil present in rock types and petrified wood in the study area. Therefore, to accomplish this study we need to answer the below question:

1. What is the characteristic of the fossil and petrified wood at Mondulkiri and Ratanakiri?
2. What is the sedimentary structure and depositional environment at there?
3. What is the paleo environment?

1.4. OBJECTIVES

The main purpose of this study is to characteristic the fossil and petrified wood which located in Mondulkiri and Ratanakiri provinces. There are three objectives in this study cases such as:

1. To identify the characteristic of fossil types and petrified wood
2. To study sedimentary structure and depositional environment
3. To characterize Paleo Environment

2. LITERATURE REVIEW

2.1. DEPOSITIONAL ENVIRONMENT

Depositional environment is part or place that possible for sediment laid on and it has certain chemical, physical biological characteristics. Sedimentary environments are interpretated by geologists based on clue within the sedimentary rock and sedimentary structures. Trace fossils, fossils and petrified wood (Deline et al., 2019). It was divided into the main depositional environments, as well as, continental environment, margin marine and marine environment.

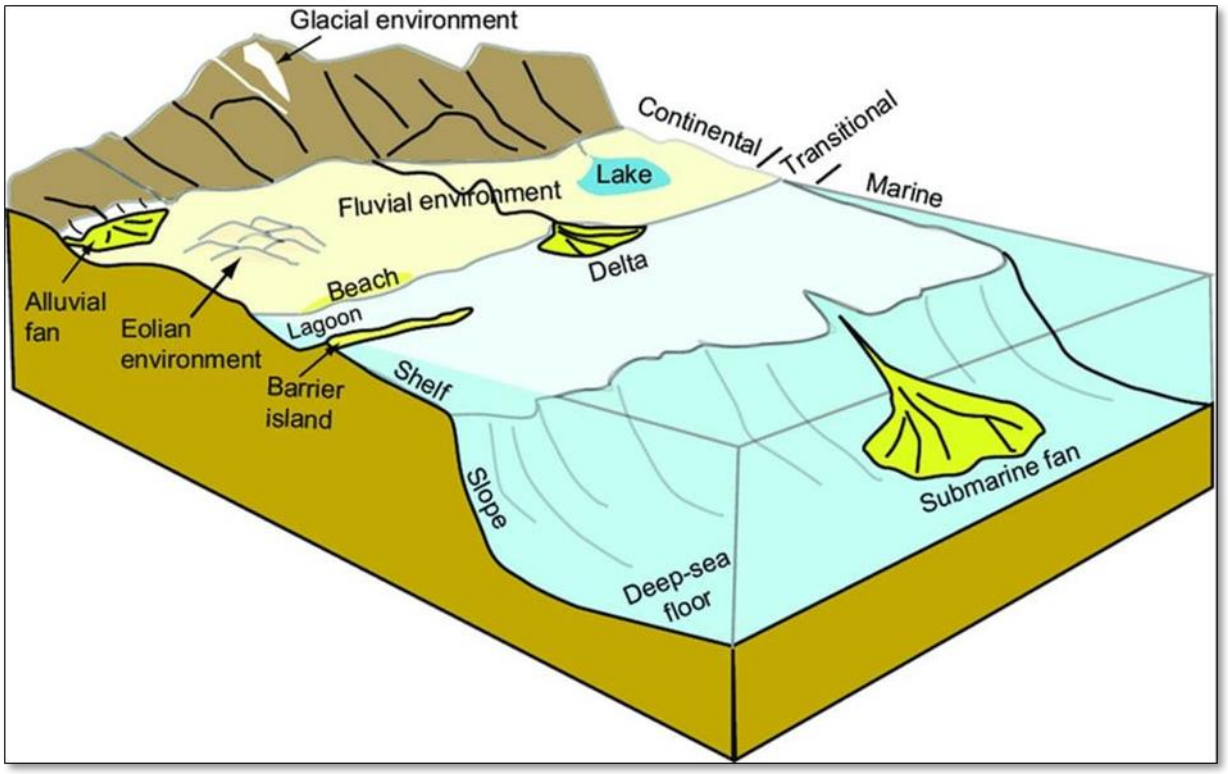


Figure 2-1.The Depositional Environment of sediment.

2.2. CARBONATE CLASSIFICATION

Carbonate rocks are composed of carbonate mineral, as well as common minerals are calcite, aragonite and dolomite. It is formed by precipitation of solution or by accumulation of fragments that remain organism. There are two types of carbonate rocks which are Limestone and Dolostone.

Limestone is a sedimentary rock composed mainly of calcite and calcium mineral and its chemical composition is CaCO3. Limestone mostly forms in the clear, calm, warm and in the shallow marine waters and usually a biological sedimentary rock which form by the accumulation of shell, coral, algal, fecal and other Origenism debris, however, it also can

form by precipitation of calcium carbonate from lake or ocean which called chemical sedimentary rock (Hobart and King, 2022).

Dolostone is composed of dolomite mineral it occurs in the certain supratidal environment, freshwater and lake present a little aragonite in Cretaceous.

The classification of carbonate rock according to Dunham classification is based on the concept of grain and mud support (Dunham, 1962).

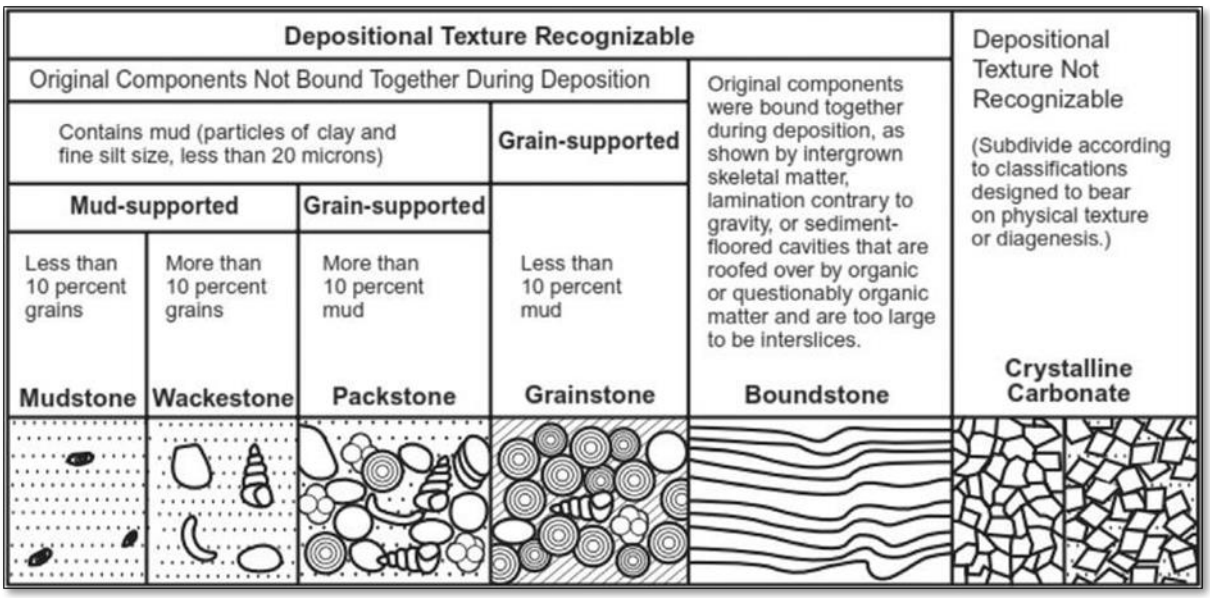


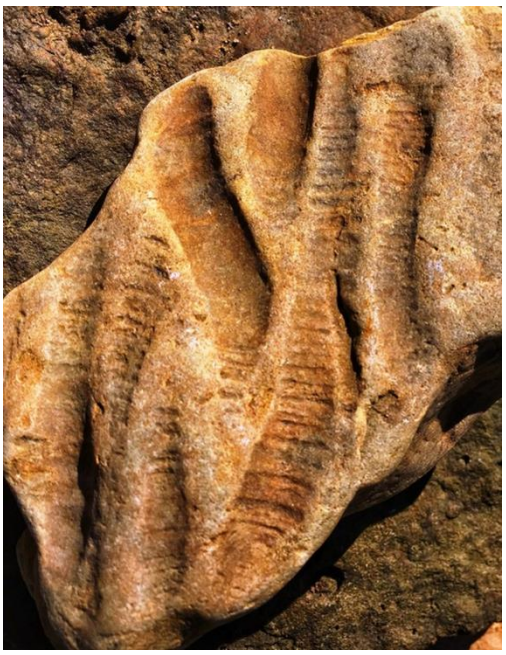
Figure 2-2.Carbonate Classification Chart, (Dunham 1962).

2.3. TYPES OF FOSSILS

A fossil is any preserved remains, impression, trace of living thing from a past geologic age. Fossils are formed when living organism died and quickly buried by sediment such as sedimentary rock. Fossils divided 4 types which are Trace fossil, Mold fossil, Cast fossil and Petrified fossil.

2.3.1. TRACE FOSSIL

Trace fossils give glimpses into how animals behaved and what their activities were like, including what they ate. Another name for trace fossils is ichno fossils. Trace fossils preserve the moment of activity of an animal and therefore indicate the habitat of the animal. trace fossils of burrows in mudstone show that the animal lived in a muddy environment. Mud accumulates in water that is very still, such as ponds, lakes, lagoons or ocean floor. So, the burrowing animal was aquatic and preferred still water. Trace fossils show the evidence of how an animal or plant interacted with its environment. They differ from body fossils--which are the preserved remains of the physical parts of an organism, such as bones and teeth.



2.3.2. MOLD FOSSIL

Mold fossils come from a process called authigenic preservation; a process which leaves a negative impression, or indent, of an organism in rock after the organism itself has deteriorated. Sand or mud covers the deceased organism and, over time, that sand or mud hardens into rock, encasing the organism. The organism continues to decay, eventually leaving only an imprint. Entire organisms, partial organisms, or even traces of the passing of organisms can leave mold fossils. Molds and casts are other types of body fossils. A mold is an imprint left by the shell of a hard skeleton on surrounding rock, such as dinosaur bones buried beneath many layers of sediment. A mold may be internal or external. An internal mold is on the underside of shell left on the surface of rock that formed when sand or mud filled the inside of the shell. An external mold is on the outside of the shell. Whenever a shell or bone breaks out of rock, it leaves an external mold behind.



2.3.3. CAST FOSSIL

Cast fossils occur when mold fossils are filled in with minerals that harden over time, creating a fossilized replica of the original organism. Water seeps through the rock surrounding the mold fossil, leaving behind minerals which fill the mold. The minerals harden, taking the shape, or the natural cast, of the mold fossil. Cast fossils are like mold fossils in that they formed, at least in part, with an imprint made in a rock or sediment. However, once the hollow molds are present, they are subsequently filled in with minerals that later harden to form solid rock through the process of permineralization. In other words, mold fossils take up negative space and cast fossils take up positive space. This casting process can use a variety of minerals including silica (often already found in plants) and pyrite (which results in shiny, metallic fossils).



2.3.4. PETRIFIED WOOD

Petrified wood is a fossil. It forms when plant material is buried by sediment and protected from decay due to oxygen and organisms. Then, groundwater rich in dissolved solids flows through the sediment, replacing the original plant material with silica, calcite, pyrite, or another inorganic material such as opal. The result is a fossil of the original woody material that often exhibits preserved details of the bark, wood, and cellular structures.



3. METHODOLOGY

3.1. RESEARCH FRAMEWORK

The aim of this study to characteristic the paleo-environment, based on Petrography by thin section, XRF (x-ray fluorescence), SEM (scanning electron microscopic) analysis. The study area was selected regarding lithological information and previous researched. After site selection, rock formation was identified by facies analysis and sampled for laboratory experiments.

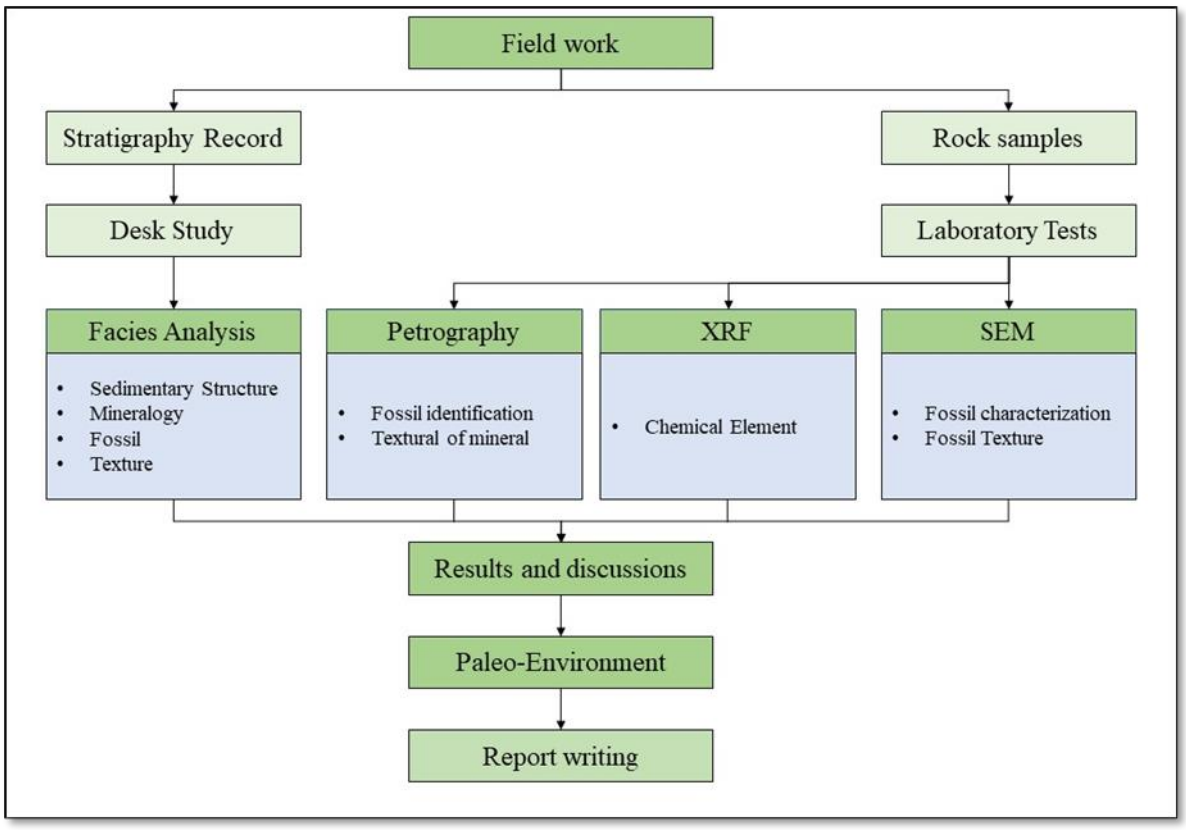


Figure 3-1. Research Flow Chart

3.2. FIELDWORK

Field work was carried out in Sre pok forest, Koh Nheaek district, Mondulkiri province and another once in Lom phat district, Ratanakiri province in order to refine the local geological map and to collect a systematic suite of representative rock samples for laboratory test. The field survey method was designed around the difficulty of research for fossil. After field survey for rock outcrop was photographed from various angles with scale, GPS location metadata. Structural measurement, character of structural feature, fossil types and mineralogy. Finally, the samples were characterized in situ before being careful get it and labeled. All the samples were taken for petrography (Thin section), XRF (X-ray fluorescence) and SEM (Scanning Electron Microscopy) analysis at as the laboratory of Faculty of Geo-Resources and Geotechnical Engineering in Institute of Technology of Cambodia.

To conduct the field observation and mapping, geologist need numerous items for the field: map, Global position system (GPS), geological hammer, compass, hand lenses, sample bags, field notebook, pen, pencil, eraser, clinometer, camera, acid bottle, first aid kit and safety with comfort clothes. Moreover, we should add mobile phone and boots for efficiently fieldwork. Observations of the sample site should take. These should include:

- GPS is utilized for point the coordinates
- Location (Latitude, Longitude and Elevation)
- Types of samples contain fossils (Ammonite, petrified wood, etc.)
- Take all the photo with outcrop
- Draw the important structure in notebook (Sketching)
- Checking all the equipment before and after finish field work



Figure 3-2. Fieldwork Equipment

3.3. STARTIGRAPGHY COLUMN MEASUREMENT

The vertical portion of stratigraphy know as a stratigraphy column is used to measure a geology sequence or the outcrop. It giving us details on outcrop lithofacies, texture and sedimentary structure to help us understand the sedimentary process and depositional setting in the studied area. Moreover, the information on the thickness of the lithology, the sedimentary structure, the geological structure, dip and strike, the composition, picture of the outcrop and sample number must be included in the sedimentary column. We also record the time, date, location, coordinate and other unique details to make it sample to identify the outcrop documented after doing geological fieldwork. GPS, Compass, Jacob staff, Hand lens, Notebook equipment for creating and documenting stratigraphic column.

3.4. HAND SPECIMEN ANALYSIS

Hand specimen is a method to identify physical properties of rock sample. Its necessary tools acquired for physical properties analysis including:

- Hand lens: used to identify on characteristic texture grain of mineral such as grain size, grain shape, grain sorting, grain rounded.
- Grain size ruler: applied to defined size grain of mineral.

Hand specimen analysis was identified at following part:

- Grain size, shape and sorting
- Color
- Mineral composition
- Structure

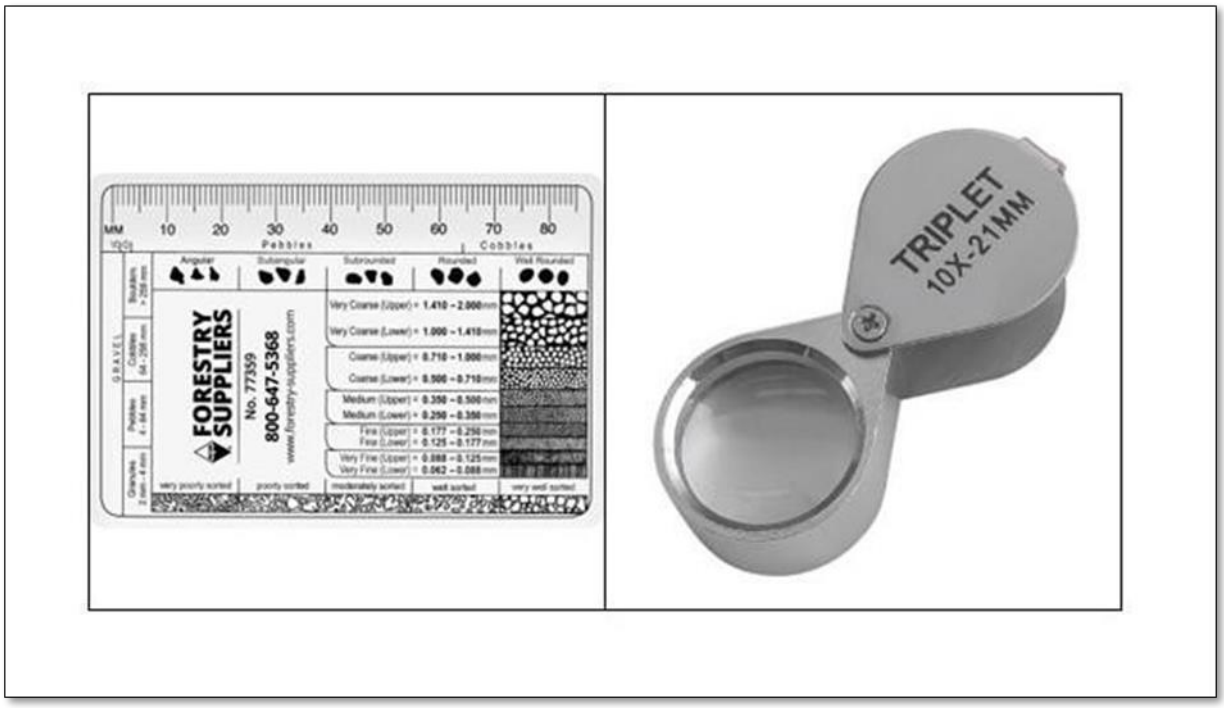


Figure 3-3. Hand specimen equipment

3.5. PETROGRAPGHY

Sample preparation as thin section requires 30 micrometers in thick of rock that use for analysis the mineral composition under microscopy. To get in thin section we need to follow the below procedure and some equipment.

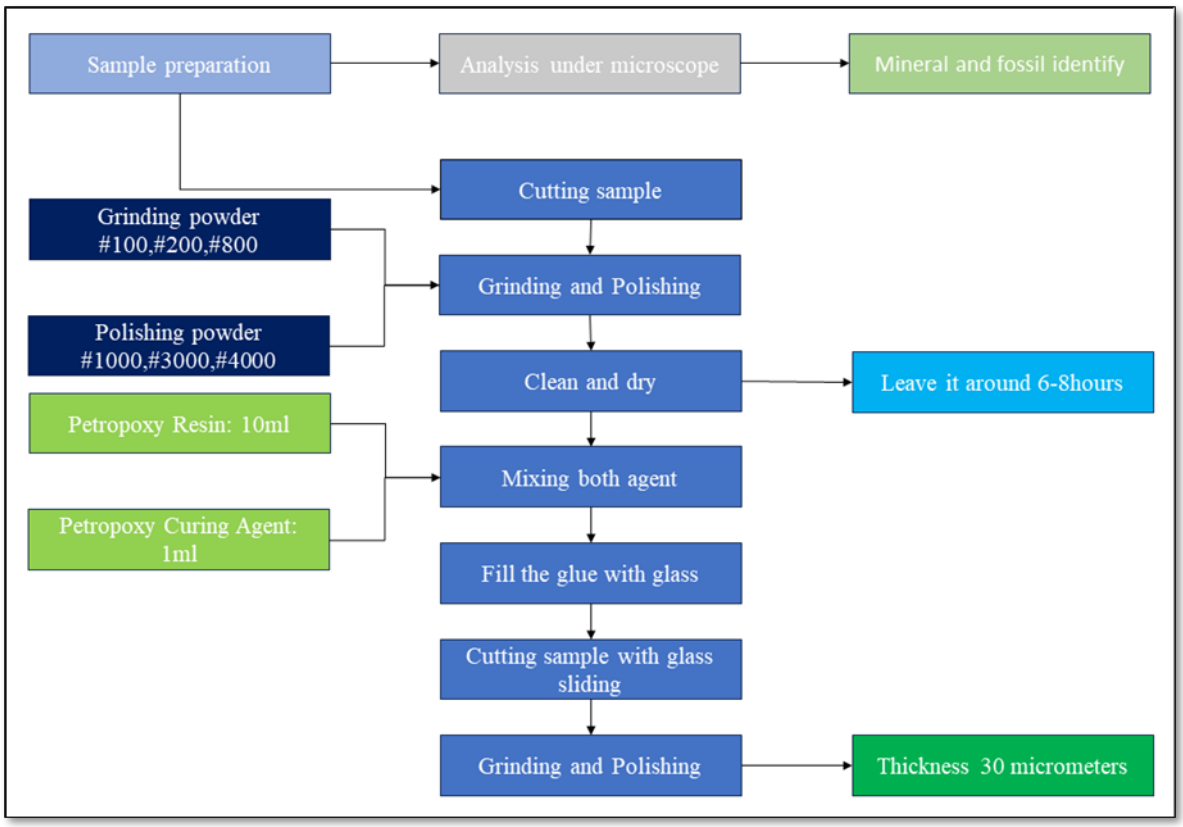


Figure 3-4. Thin Section procedure.

- Sample should be cut with large- and small-scale machine to get appropriate glass slide size
- Grind the sample with grinding machine from #100, #200 and #800. Then polish it with grinding plate from #1000, #3000 and #4000 for obtain sample's smooth surface.
- Place it on hot plate around 6-8 hours to make sure the sample is dry before sticking with a glass slide
- Mix the ration of Petropoxy resin 10ml, Petropoxy curing agent 1ml. after that fill the mixer on the surface of the sample and keep it on hot plate around 6-8 hours until it dry and fit with the glass
- Finally, the thin section sample is finished for analyze under microscope.

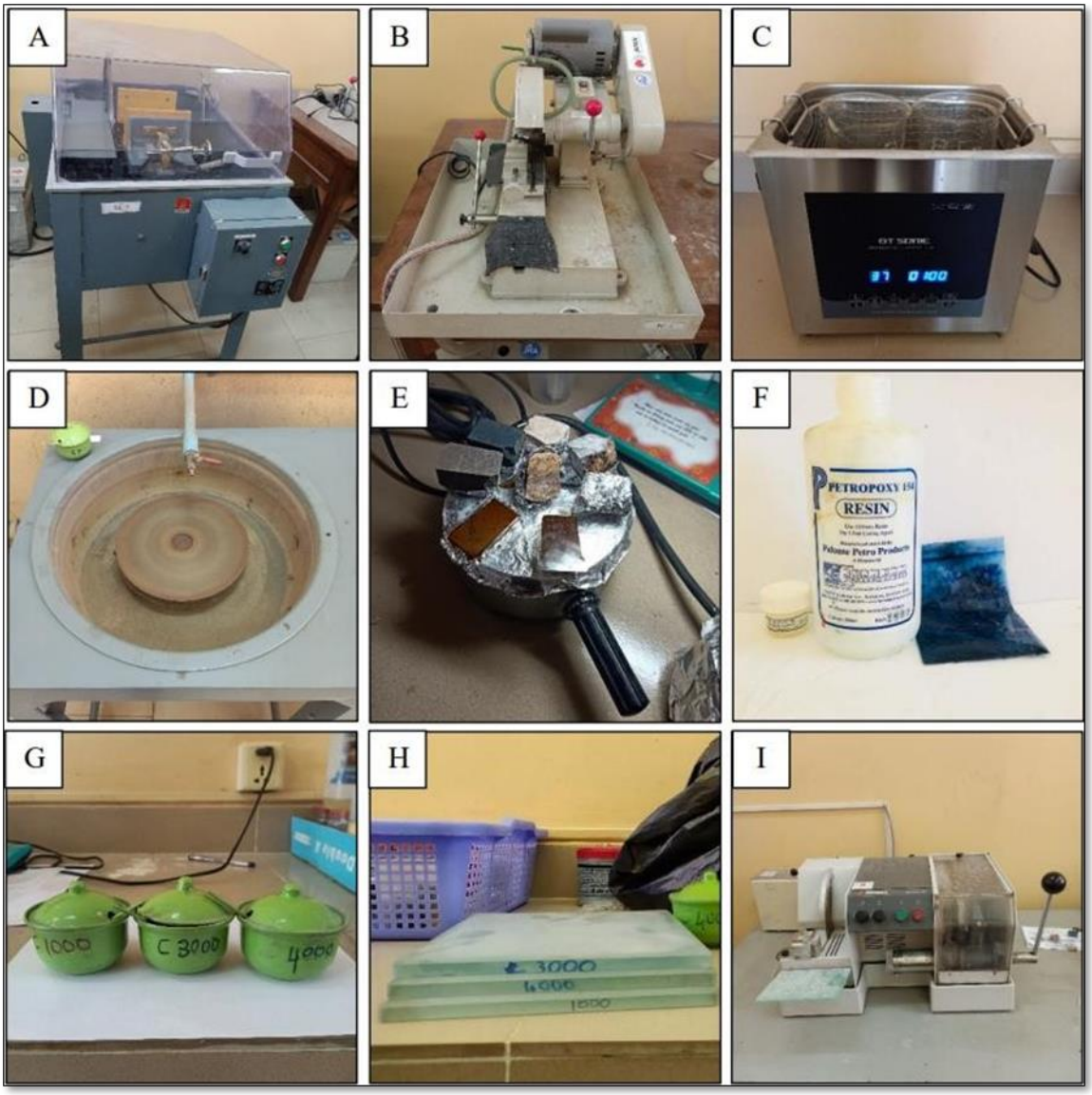


Figure 3-5. Thin Section Equipment

3.6. X-RAY FLUORESCENCE (XRF)

X-Ray Fluorescence (XRF) is analyzed to determine the major element and oxide elements content in the rock sample. However, before testing, the rock sample was dried using an oven at 105° for a day (24hours). To complete this method, we need to prepare sample with the equipment and procedure as below:

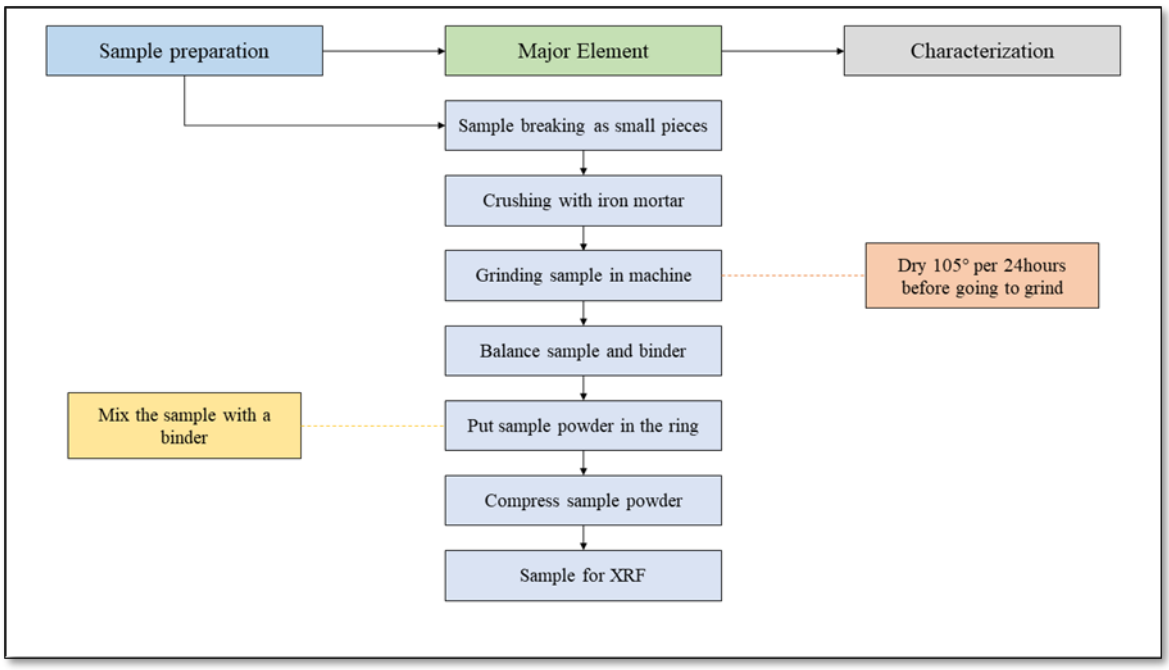


Figure 3-6. XRF procedure

i. Break the rock sample with a hammer to get small pieces.
ii. Crush the pieces of rock with an iron mortar.
iii. Grind that sample after crushing with either agate mortar or milling machine to get the fine powder (less than 50 µm).
iv. Weight the powder sample 5.4g and binder 1g, then mix with each other to homogenously powder.
v. Put the powder into PVC ring or aluminum ring.
vi. Pressing the ring with sample around 18 to 20 T force or MPa.
vii. Sample after pressing should be clean the surrounding the ring.

3.7. SCANNING ELECTRON MICROSCOPE (SEM)

Scanning Electron Microscopy (SEM) is used to locate and identify minerals (Bohor and Hughes, 1971). SEM operates by focusing on an electron beam with high intensity. When a beam electron strikes a sample, then detect to provide a picture depending on the surface tomography. Additionally, there includes information the structure surface about the feature that was seen in the SEM micrograph.



Figure 3-7. Scanning electron Microscopy (SEM/EDX)

The procedure of SEM-EDS: follow the step as thin section until last step then take it polish with polishing plate, After that take that sample for coating with carbon.

Noted: sample surface should be smooth before coating.

4. RESULTS

4.1. HAND SPECIMEN

Hand specimen giving the information for identify it by naked eyes which can know grain size, grain shape, grain sort, grain support, mineral composition, texture, fossil types (trace fossil, body fossil, Mold fossil).



Figure 4-1. Hand Specimen

4.2. STRATIGRAPGHY RECORD

Stratigraphy measure can divide into lithofacies which can know the structures on the them such as fault, fold, lamination, bedding, fracture, strike and dip. from the outcrop and field observation, Nevertheless, all that formation of those structure giving a result where it forms, transportation, erosion, replacing…, etc.

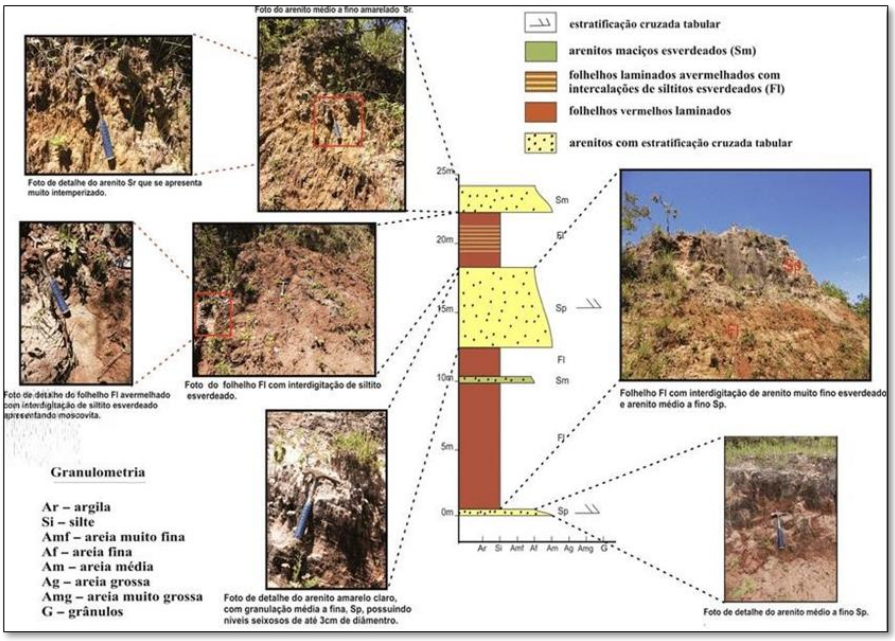


Figure 4-2. Stratigraphy Analysis.

4.3. THIN SECTION

Petrography used for examine the rock's mineral content. For an Ammonite fossil substance in shells of three Jurassic. The shell of one ammonite with partly retained internal structure has been replaced with quartz, carbonate bearing apatite (CaF), and newly formed aragonite.

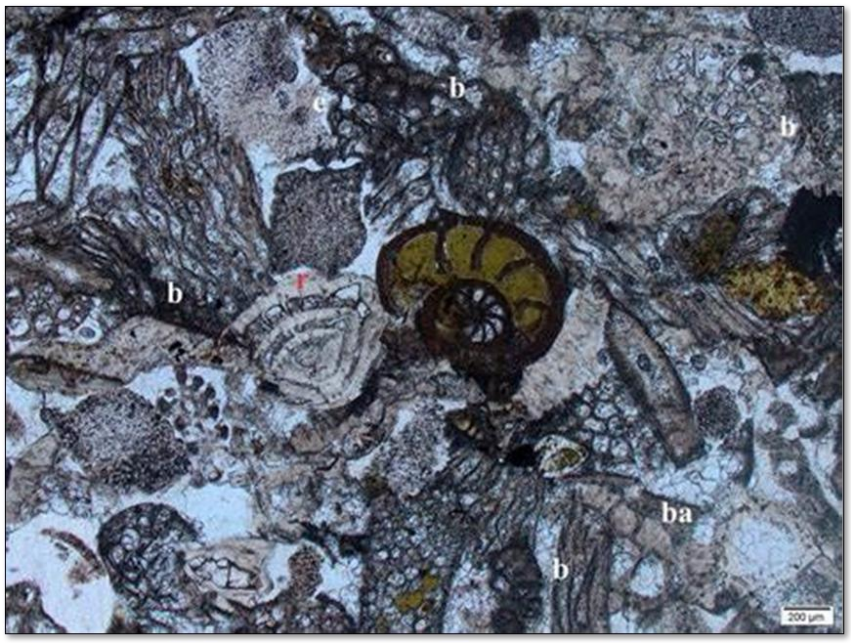


Figure 4-3. limonite under microscope

4.4. X-RAY FLUORESCENCE (XRF)

X-Ray Fluorescence (XRF) is analyzed to determine the major element and oxide elements content in the rock sample. The major of an element that are contained such as: CaO, SiO2, Fe2O3, Al2O3, MgO which are present as oxide elements in percentages that can be identify in the depositional environment.

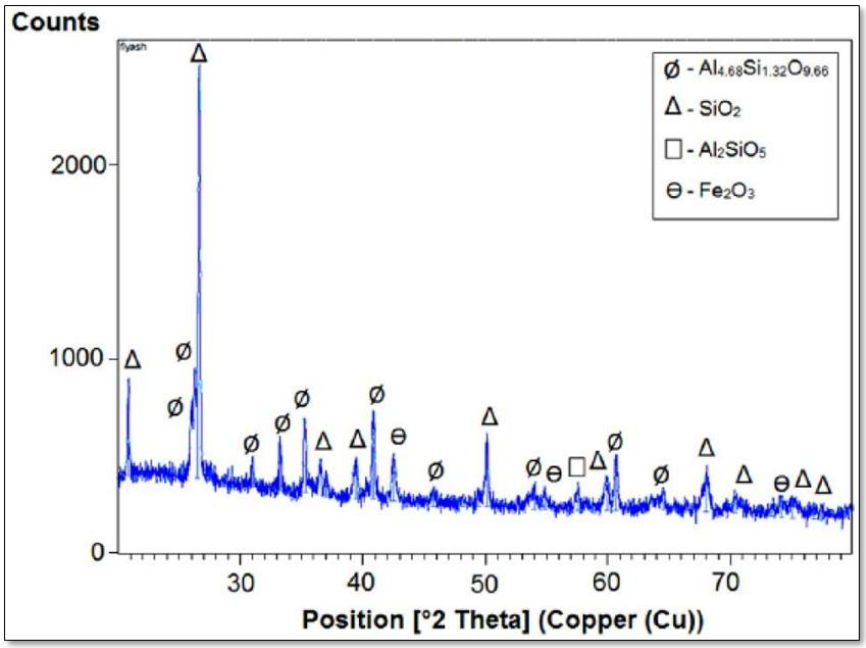


Figure 4-4. XRF analysis

4.5. SCANNING ELECTRON MICROSCOPY (SEM)

SEM is an important tool for identify the morphology of microfossil, which can detect the structure surface in nano scale incase for more conformity with shape and texture with thin section.

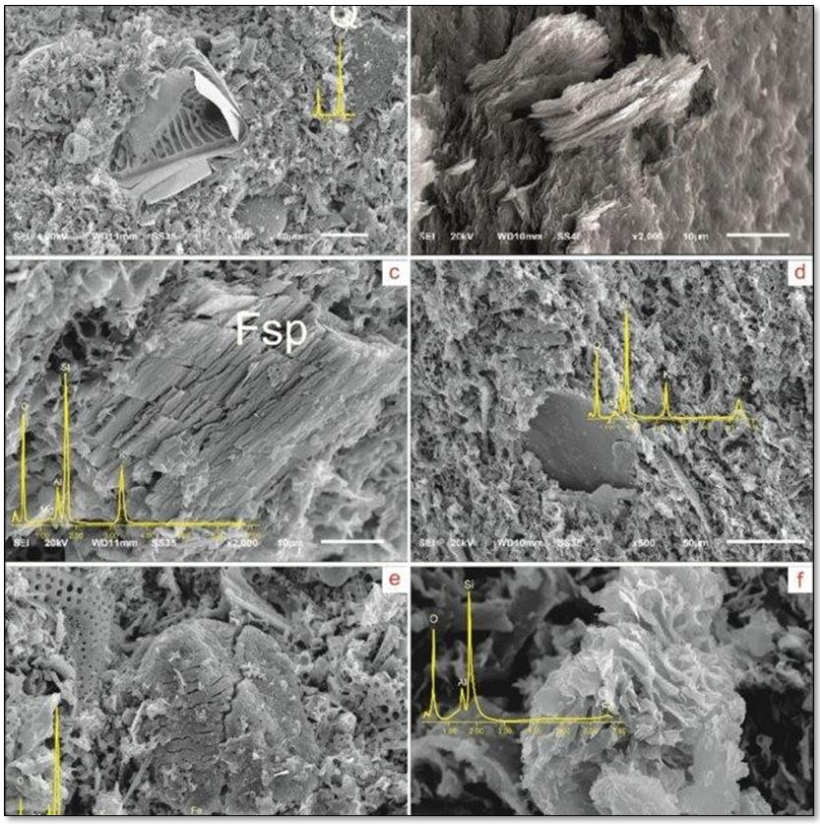


Figure 4-5. SEM analysis result.

5. CONCLUSION

All of the advantages method above is the main key for analysis of paleo-environment and stratigraphy record.

- Petrography which can identify the mineral composition fossil content in case for giving it rock type and the fossil's name.
- X-Ray Fluorescence (XRF) can detect the major element for plotting the depositional environment where it from.
- Scanning Electron Microscopy (SEM) which detect in nano scale for clarifying with structure and element also.

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